AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Introduction



Workshop goals

By the end of the workshop you will be able to:

- wrangle and visualise time series data in R
- produce forecasts with statistical models and time series foundation models
- evaluate forecast accuracy and choose between models

Project lead and Instructors



Bahman Rostami-Tabar

Professor of Data Driven Decision Making and Director of the Data Lab for Social Good, Cardiff Business School. Founder of the Forecasting for Social Good (F4SG) community and leader of the AFRICAST project.

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Mitchell O'Hara-Wild

Time series researcher and R package developer from Monash University (Australia).
Developer of the time series packages (`fable`, `tsibble`, `ggtime`, ...) used throughout this workshop.

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Instructors



Abiodun Egwuenu

Instructor and workshop facilitator. Medical doctor and field epidemiologist. PhD candidate at Charité – Universitätsmedizin Berlin modelling antimicrobial resistance (AMR). Led Nigeria's 2022/23 mpox response.



Harsha Halgamuwe Hewage

Researcher, Data Lab for Social Good, Cardiff Business School. Forecasting and decision support for public health supply chains; researches time series foundation models.



Mustafa Aslan

PhD researcher, Data Lab for Social Good, Cardiff University. Probabilistic forecasting and optimisation for hospital patient flow. Developer of the `peshbeen` forecasting package.

Organisers

Coordinator

- Henry Kissinger Ochieng

Planning team

- Mkajuma Charity
- Kennedy Mung'are
- José Thiéry Hagbe

Mentors

- Samson Oluwafemi
- Oluchukwu Aghadi
- Lumumba Victor
- Adamu SAIDU Saleh
- Akoeugnigan Idelphonse SODE
- AKOUNDA Badjibassa
- Caroline Mugo
- Winnie Chacha
- Matabel Odin

Assumptions

- Not an R introduction: assumes comfort with R, an IDE (RStudio or Positron) and the tidyverse.
- Not a statistics course: assumes familiarity with means, standard deviations, quantiles, regression, the normal distribution, likelihood, etc.
- Not a theory course: no derivations, just tools for time series and forecasting, when to use them and how to use them well.

Approximate outline

Session	Topic	Chapter
1	Basics of time series and data structures	2
1	Time series patterns and basic graphics	2
2	Computing and visualising features	4
2	Transformations and decompositions	3
2	Basic modelling / forecasting	1,5
3	Forecasting with regression	7,10
3	ARIMA models	9
4	Exponential smoothing	8
4	Basic training and test accuracy	5
5	Cross-validation and residual diagnostics	5
5	Time series foundation models	–

workshop.f4sg.org/africast

Logistics

Each session (UTC+3)

- Lecture 1: 10:00–11:30
- Break: 11:30–11:45
- Lecture 2: 11:45–13:15
- Exercises: 14:00–17:00 (on your own)
- Mentorship: 17:00–18:00

Materials and support

- Workshop website
- Mentorship sessions and Q&A
- Google Classroom

Install required packages

Install before session 1:

```
install.packages(c(
  "tidyverse",
  "fpp3",
  "GGally",
  "ggtime",
  "astsa"
))
```

Check it works:

```
library(fpp3)
```

References

Hyndman, R. J. & Athanasopoulos, G. (2021) *Forecasting: principles and practice*, 3rd ed.

OTexts.org/fpp3/

- Free and online
- Data sets in associated R package
- R code for examples

Kolassa, S., Rostami-Tabar, B. & Siemsen, E. (2023) *Demand forecasting for executives and professionals*, 1st ed.

dfep.netlify.app

- Free and online
- Not an in-depth technical book
- Mindset behind forecasting
- Overview of forecasting methods and processes

Community

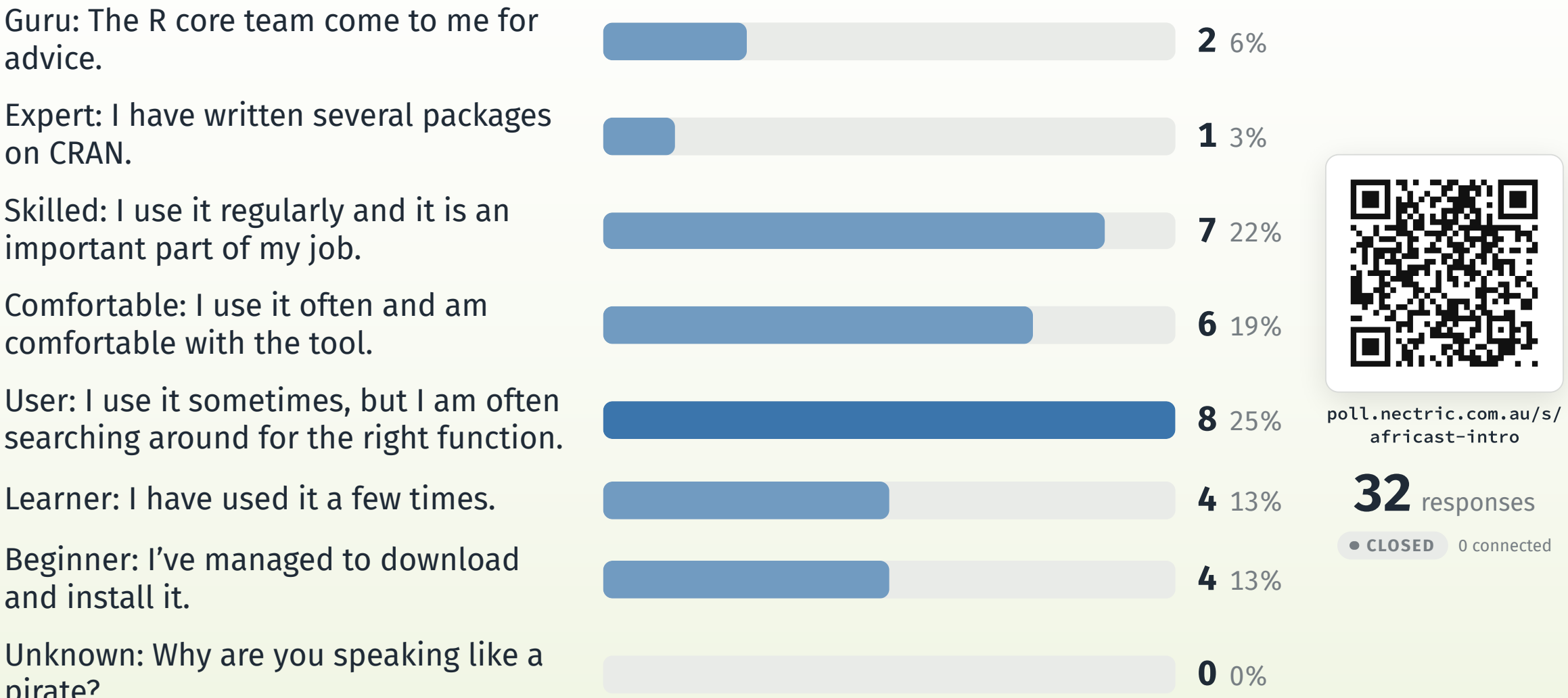
International Institute of Forecasters (IIF)

- Certificate provided by IIF: attend all lectures and exercise sessions to receive it.
- Nonprofit founded in 1982, dedicated to developing and furthering the generation, distribution and use of knowledge on forecasting.

Forecasting for Social Good (F4SG) committee

- Website: f4sg.org
- Follow us on LinkedIn
- Use #AFRICAST, #F4SG if you post on social media about the workshop

Poll: How proficient are you in using R?



Poll: How experienced are you in forecasting?

